

Abrar Shaikh

Embedded Software Engineer

✉ abraratjob@gmail.com

☎ 7021906137

📍 Mumbai-88

💻 [website](#)

EDUCATION

- Bachelor of Technology
Electrical Electronic Engineering

Manipal University

📅 2017 - 2021

🎓 8.1 CGPA

📍 Manipal, Karnataka

- High School

🎓 85% in PCM

- Secondary School

🎓 91%

SKILLS

- Embedded C, C++
- ARM Cortex A, ARM ASM
- Tarmac Trace, BDM Debugging
- Python, Perl, tcsh
- AutoSAR
- Rhapsody Design
- CAN, UDS, XCP, CAPL
- SPI, UART, ADC
- Verilog HDL

TOOLS

- Linux, RTOS, Git
- Armclang compiler toolchain
- Eclipse IDE, IAR, QAC
- CANoe, CANalyzr, CANstress
- DOORS-DNG, ALM, Jira
- Pico scope, logic analyser
- Xilinx Vivado

INTEREST

- Embedded HW-SW Co-Design
- Advance Optimization Techniques
- Analytical Mathematics

COURSES

- Embedded Linux YOCTO: [Udemy](#)
- MATLAB Simulink: [MathWorks](#)
- Unix Essential Training: [LinkedIn](#)

A dedicated, results-oriented Embedded software engineer with more than **4.8+ years** of experience, proficient in **Embedded C bare-metal programming**, and a self-motivated, eclectic learner, seeking immediate challenging role to leverage my strong analytical skills, solid experience in **SDLC**, and passion for problem-solving to contribute effectively to a dynamic team.

WORK EXPERIENCE

ARM

Embedded Software Engineer

📅 Oct 2024 – June 2026

📍 Bangalore

- Actively contributing to development of HAL-Hardware Abstract software and infrastructure for Cortex-A **v10a, v9a** Architecture verification.
- Done upto **78%** of runtime optimisation in Bring-Up test in RTL simulation.
- Achived **75%** of compile time optimisation of Test Library Build flow.
- Maintaining BOOT, Drivers code and verification Test Abstraction APIs.
- Done code optimisation in Boot to save **~8KB** of space in final ELF.
- Developed strong Debugging skill of CPU execution flow using ARM **TARMAC Trace** and disassembly, ELFs.
- Actively collaborated with Partner and Suite teams under stringent timeline.

American Megatrends - AMI

Senior Software Engineer

📅 July 2024 – Sept 2024

📍 Remote

- ARM Cortex-M, Embedded Linux, YOCTO, GIT, JIRA

BOSCH BGSW

Software Engineer

📅 September 2021 – May 2024

📍 Bangalore

- Responsible for Embedded application software development for Exhaust gas NOx measurement with *Sensor Controller Unit-SCU* (NXP s12z 16bit MCU)
- Involved in developing End of Line (EOL) Production SW and various layers in BSW like MCAL and CDD.
- Maintained quality code with MISRA-C standard using Static code analyser QAC.
- Contributed to multiple stages of V-model compliance with ASPICE.
- Performed HIL Testing, Verification and Validation using XCP and UDS.
- Automated several Test Cases, and saved **100 of hrs** in manual effort.

Sikkim Manipal institute of Technology

Academic Research

📅 July 2019

📍 SMIT - Sikkim

- Worked on Advance Optimization Techniques like integrated Fuzzy-COPRAS methodology for multi criteria-based Decision-making Algorithms.
- Publication: [Taylor and Francis](#)

Personal Projects

RISC-V based GP-IO Design - [GitHub](#)

📅 Jan 2024

- Implemented 4x4 keypad driver in C with inline Assembly
- Assembly Dump was used to generate RTL HDL code.
- RTL functional simulation was done using GTK-wave.
- Yosys toolchain was used to synthesise RTL HDL code with Skywater Foundry 130nm based standard cells.

FPGA based Image processing - [GitHub](#)

📅 June 2021

- Implemented 2D Convolution RTL design using Verilog for Image Processing applications like *Blur, sharpen* and *Edge detection*.
- Functional verification was done using Verilog-based testbench in Vivado.